LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	13
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Types of brake unit, general design requirements and brief description of a typical brake unit, friction discs and pads, brake master cylinder
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Types of brake unit, general design requirements and brief description of a typical brake unit, friction discs and pads, brake master cylinders
5. Specific Objectives	23.1.3.1 Trainee will be able to describe typical alighting gears of an aircraft and its construction.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Examination of brake unit components, safety precautions regarding hydraulic pressure, instructor supervision during component identification
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the landing gear braking system?
(b) Name two general design requirements for aircraft brakes.
(c) How does the master cylinder initiate the braking action?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Types of brake units • General design requirements • Typical brake unit description • Friction discs and pads • Brake master cylinder operation	L & Ds	PPT, IB	15 min
(b)	• Hydraulic pressure generation • Friction material selection • Disc and pad assembly • Master cylinder piston movement • Brake fluid flow path	L & Ds	PPT, PDF	20 min
(c)	• Component identification • Safety inspection points • Wear limit assessment • Assembly sequence review • Troubleshooting basics	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between master cylinder pressure and brake disc friction.
(b) Identify three critical design requirements for high-performance aircraft brakes.
(c) Describe the role of friction pads in converting hydraulic energy to stopping force.

12. Home Assignment and Follow Up	Prepare a brief note on the operating principle of anti-skid devices to be discussed in the next lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Anti-skid devices, purpose and operating principle of anti-skid unit.	Refer to the master syllabus reference list	LP 14

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min